Part I: Multiple choice questions (20 marks)

Q1: If V is a 6-Dimensional vector space and $v_1, v_2, \dots, v_m$ is a basis for V, then one of the following is true about m.

A) $m = 6$ B) $m \geq 6$ C) $m \leq 6$ D) $m \neq 6$

Q2: The inverse of $A = \begin{bmatrix} a & 0 & 0 \\ 0 & 2a & 0 \\ 0 & 0 & a \end{bmatrix}, a > 0$ is:

A) $\begin{bmatrix} a & 0 & 0 \\ 0 & 2a & 0 \\ 0 & 0 & a \end{bmatrix}$ B) $\begin{bmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{2a} & 0 \\ 0 & 0 & \frac{1}{a} \end{bmatrix}$ C) $\frac{1}{2a^2}\begin{bmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{2a} & 0 \\ 0 & 0 & \frac{1}{a} \end{bmatrix}$ D) Not invertible

Q3: $\begin{vmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & d & 1 \end{vmatrix} =$

A) $abcd$ B) a C) $-c$ D) c

Q4: Which of the following vectors is an eigenvector for the matrix $\begin{bmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{bmatrix}$

A) $\begin{bmatrix} \frac{-2}{5} \\ 1 \end{bmatrix}$ B) $\begin{bmatrix} 1 \\ \frac{-2}{5} \end{bmatrix}$ C) $\begin{bmatrix} \frac{-2}{3} \\ 1 \end{bmatrix}$ D) $\begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix}$

Q5: Which of the following statements are true,

A. If A is 2×2 matrix, then $\{u; Au = 3u\}$ is subspace of $\mathbb{R}^2$,

B. If A is 2×2 matrix, $\left\{u; Au = \begin{bmatrix} 2 \\ 0 \end{bmatrix}\right\}$ is subspace of $\mathbb{R}^2$,

A) A only B) B only C) Both A and B D) Neither A nor B

Q6: Let $V = \{All\ symmetric\ matrices\}$, **then**

A) V is vector space B) V is not a vector space C) $dim\ V = \frac{n}{2}$ D) $A + C$

Q7: One of the following is not linear transformation:

A) $T(p(x)) = lnp(x)$ B) $T(p(x)) = xp(x)$ C) $T(p(x)) = \int p(x)\,dx$ D) $T(p(x)) = \frac{d}{dx}(p(x))$

Q8: Let $T: P_2 \to P_3, T(p(x)) = xp(x)$, **which of the following belongs to Range** T

A) $x + x^3$ B) $1 + x$ C) $3 - x$ D) $3 - x^3$

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Part two: True or False questions: (20 marks)

Q1: Let $D: P_n \to P_{n-1}$ be the differentiation transform $D(p(x)) = p'(x)$, then $KerT$ is all the linear polynomials.

(TRUE\ FALSE)

Q2: If $T(c_1v_1 + c_2v_2) = c_1T(v_1) + c_2T(v_2)$, then T is a linear transformation. (TRUE\ FALSE)

Q3: If $T: V \to W$ is a linear transformation then $Ker\ T$, is not a vector space. (TRUE\ FALSE)

Q4: The linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + y, x - y)$ is not one to one.

(TRUE\ FALSE)

Q5: If 0 is an eigenvalue of a matrix A, then A^2 is singular. (TRUE\ FALSE)

Q6: If the characteristic polynomial of a matrix A is $p(\lambda) = \lambda^2 + \lambda + 1$ then A is invertible.

(TRUE\ FALSE)

Q7: If λ is an eigenvalue for an invertible matrix A then $\frac{1}{\lambda}$ is an eigenvalue for A^{-1}.

(TRUE\ FALSE)

Q8: The dimension of all diagonal $n \times n$ matrices is $2n$. (TRUE\ FALSE)

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